

A proof of Kurkov's conjecture for A000670

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Abstract

The Fubini numbers $a(n)$ (OEIS A000670), which count the ordered set partitions of an n -element set, satisfy a conjecture of Mikhail Kurkov (2018) recorded on the entry: for $n > 0$,

$$a(n) = \sum_{k=0}^{2^{n-1}-1} A284005(k), \quad a(0) = 1,$$

where $A284005$ (OEIS A284005) is defined by $A284005(0) = 1$ and $A284005(m) = (1 + \text{wt}(m)) A284005(\lfloor m/2 \rfloor)$, with wt the binary digit sum. We give a short, elementary, and self-contained proof. The key is a refined statement: for each of the 2^{n-1} subsets $S \subseteq [n]$ containing 1, the number of ordered set partitions of $[n]$ whose set of block minima equals S is exactly $A284005(k)$ for the index k that encodes S . Summing over S proves the conjecture. The refinement is verified exhaustively for $n \leq 8$ (598,444 ordered set partitions across 255 block-minima classes, 0 mismatches).

1 Introduction

Let $a(n)$ denote the number of *ordered set partitions* of $[n] = \{1, \dots, n\}$: partitions of $[n]$ into nonempty blocks together with a linear order on the blocks. These are the Fubini (or ordered Bell) numbers, OEIS entry A000670 [1], with $a(0) = 1$ and initial values 1, 1, 3, 13, 75, 541, 4683, ...

The entry A000670 carries the following formula, contributed by Mikhail Kurkov on July 8, 2018, and still labelled a conjecture:

$$a(n) = \sum_{k=0}^{2^{n-1}-1} A284005(k) \quad (n > 0), \quad a(0) = 1, \quad (1)$$

where $A284005$ is the OEIS entry A284005 [2], defined by

$$A284005(0) = 1, \quad A284005(m) = (1 + \text{wt}(m)) A284005(\lfloor m/2 \rfloor), \quad (2)$$

and $\text{wt}(m)$ is the number of 1's in the binary expansion of m . The purpose of this note is to prove (1).

The proof is elementary and combinatorial. We first unwind (2) into a product over the binary digits of k (Section 2); this lets us index the summands of (1) by subsets of $[n]$ containing 1. We then prove a refinement of (1) (Section 3): each summand counts exactly the ordered set partitions with a prescribed set of block minima. Because every ordered set partition of $[n]$ has a unique set of block minima, and 1 is always a block minimum, summing the refinement over all 2^{n-1} admissible subsets yields (1).

To our knowledge (1) was open before this work: it is stated as a conjecture on A000670, and a dated literature and web search (July 2026) turned up no proof.

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2 The product formula and its index set

Iterating (2) strips the binary expansion of m from the least significant bit. Writing m in binary as $b_1 b_2 \cdots b_L$ (most significant bit first, $b_1 = 1$), the successive arguments $\lfloor m/2 \rfloor, \lfloor m/4 \rfloor, \dots$ are the prefixes $b_1 \cdots b_{L-1}, b_1 \cdots b_{L-2}, \dots$, and each contributes the factor $1 +$ (its digit sum). Hence

$$A284005(m) = \prod_{i=1}^L (1 + w_i), \quad w_i := b_1 + b_2 + \cdots + b_i, \quad (3)$$

the same product formula recorded on A284005. A prefix of digit sum 0 contributes a factor $1 + 0 = 1$; equivalently, *prepending a 0 to the binary string does not change the value*. (Here $A284005(0)$ is the empty product 1.)

Fix $n > 0$. Every index k with $0 \leq k \leq 2^{n-1} - 1$ can be written, after left-padding with zeros, as a string of exactly $n - 1$ bits $\mathbf{b} = b_1 b_2 \cdots b_{n-1}$ (most significant first). By the padding invariance just noted, (3) gives

$$A284005(k) = \prod_{i=1}^{n-1} (1 + w_i), \quad w_i = b_1 + \cdots + b_i, \quad (4)$$

for every such k , using its $(n - 1)$ -bit padded expansion. As k ranges over $\{0, \dots, 2^{n-1} - 1\}$, the string $\mathbf{b}$ ranges over all of $\{0, 1\}^{n-1}$.

Definition 2.1. For $\mathbf{b} = b_1 \cdots b_{n-1} \in \{0, 1\}^{n-1}$ put

$$M(\mathbf{b}) = \{1\} \cup \{i + 1 : b_i = 1\} \subseteq [n].$$

The map $\mathbf{b} \mapsto M(\mathbf{b})$ is a bijection from $\{0, 1\}^{n-1}$ onto $\{S \subseteq [n] : 1 \in S\}$: bit b_i records whether $i + 1 \in S$, and 1 is always included. There are 2^{n-1} such subsets, matching the range of k .

3 The refined theorem

Theorem 3.1. *Let $n > 0$ and $\mathbf{b} \in \{0, 1\}^{n-1}$. The number of ordered set partitions of $[n]$ whose set of block minima equals $M(\mathbf{b})$ is exactly $\prod_{i=1}^{n-1} (1 + w_i)$, where $w_i = b_1 + \cdots + b_i$.*

Before proving Theorem 3.1 we record how it settles the conjecture.

Corollary 3.2. *For every $n > 0$, $a(n) = \sum_{k=0}^{2^{n-1}-1} A284005(k)$.*

Proof. Every ordered set partition of $[n]$ has a well-defined set of block minima, which necessarily contains 1 (the block containing 1 has minimum 1). Thus the ordered set partitions of $[n]$ are partitioned according to their block-minima set S , and S ranges over exactly $\{S \subseteq [n] : 1 \in S\}$. By Theorem 3.1 and (4), the class with minima set $S = M(\mathbf{b})$ has size $A284005(k)$, where k is the integer with padded binary string $\mathbf{b}$. Summing over the 2^{n-1} admissible S , equivalently over $k = 0, \dots, 2^{n-1} - 1$, gives $a(n) = \sum_k A284005(k)$. The case $n = 0$ is the boundary value $a(0) = 1$. $\square$

Proof of Theorem 3.1. We build the ordered set partitions of $[n]$ by inserting the elements $1, 2, \dots, n$ one at a time in increasing order, tracking the ordered list of blocks. We use one observation throughout: *because elements are inserted in increasing order, an element is the minimum of its block if and only if it opened that block* — every element that joins an already existing block is larger than that block's opener, hence not its minimum.

Start with element 1, which necessarily opens the single block ($\{1\}$); after this step there is $m_1 = 1$ block. Now suppose we have an ordered set partition of $[i]$ with m_i blocks, and we insert element $i + 1$ ($1 \leq i \leq n - 1$). We want element $i + 1$ to be a block minimum precisely when $b_i = 1$ (so that, by Definition 2.1, $i + 1$ enters the minima set exactly when $b_i = 1$). By the observation above:

- if $b_i = 1$ (element $i + 1$ must be a block minimum), it must *open a new singleton block*, which can be inserted in any of the $m_i + 1$ positions of the ordered list of blocks;
- if $b_i = 0$ (element $i + 1$ must not be a block minimum), it must *join one of the m_i existing blocks*.

The number of blocks after step i is $m_{i+1} = 1 + w_i$: it starts at $m_1 = 1 = 1 + w_0$ (with $w_0 = 0$) and increases by one exactly when $b_i = 1$, i.e. $m_{i+1} = m_i + b_i$ and $w_i = w_{i-1} + b_i$. Consequently the number of choices at step i is, in both cases, exactly $1 + w_i$:

$$b_i = 1 : \quad m_{i+1} = (1 + w_{i-1}) + 1 = 1 + w_i \quad (w_i = w_{i-1} + 1); \quad b_i = 0 : \quad m_i = 1 + w_{i-1} = 1 + w_i \quad (w_i = w_{i-1}).$$

Therefore the number of insertion histories realising the prescribed signature $\mathbf{b}$ is $\prod_{i=1}^{n-1} (1 + w_i)$.

It remains to see that these insertion histories are in bijection with the ordered set partitions of $[n]$ whose block-minima set is $M(\mathbf{b})$. Distinct choices at any step produce distinct ordered set partitions, so distinct histories give distinct outputs. Conversely, every ordered set partition P of $[n]$ arises from exactly one history: deleting the element n (and deleting its block if n was a singleton) recovers an ordered set partition of $[n - 1]$ together with the last choice made, and this reconstruction is deterministic; induction on n gives a unique history for P . Under this bijection the signature of the history recording which insertions opened a new block is exactly the indicator of the block minima of P among $2, \dots, n$ — since removing the maximum of a block never changes that block's minimum — so the histories with signature $\mathbf{b}$ correspond precisely to the ordered set partitions with block-minima set $M(\mathbf{b})$. This proves the count. $\square$

4 Verification

The identity (1) was checked numerically for $n = 1, \dots, 20$ by computing $a(n)$ from the Fubini recurrence and independently from the right-hand side of (1) via (2); the two agree.

Theorem 3.1 was verified *exhaustively* for $n = 1, \dots, 8$ by generating every ordered set partition of $[n]$ (there are

$$1 + 3 + 13 + 75 + 541 + 4683 + 47293 + 545835 = 598,444$$

of them in total across $n = 1, \dots, 8$), grouping them by block-minima set, and comparing the class sizes to $A284005(k)$ over all 255 minima-set classes ($1 + 2 + \dots + 128$). There were 0 mismatches. A mutation-testing harness, which perturbs the $A284005$ recurrence and the block-minima encoding, confirms that the check is a genuine discriminator: every mutant is rejected.

These computations corroborate but do not replace the proof of Section 3, which is elementary, finite, and self-contained.

Acknowledgments

This note, including all of its mathematical content, was produced by an artificial-intelligence workflow, and we believe readers and referees should be able to take this fact into account. The

conjecture was selected, attacked, and proved by language-model agents (Anthropic’s Claude family) running in Demonstrandum, an orchestrated, verification-first multi-agent research pipeline. The definitions of A000670 and A284005 were pinned as verbatim quotations of the OEIS entries before any proof attempt; the proof was then subjected to an adversarial audit in which a model from a different family (OpenAI’s GPT-5.5, accessed through the Codex CLI with code-execution access) acted as a hostile referee, prompted to find counterexamples and gaps rather than to confirm, and wrote its own clean-room verification code. No human mathematician contributed mathematical content; the human role was limited to operating the pipeline. Correctness rests on the elementary proof printed above, intended to be checkable line by line in the ordinary way; the machine checks of Section 4 are corroboration, not a substitute.

References

- [1] OEIS Foundation Inc., Entry A000670, *Fubini numbers* (ordered set partitions), in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A000670> (accessed July 2026). Conjecture of M. Kurkov, July 8, 2018.
- [2] OEIS Foundation Inc., Entry A284005, in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A284005> (accessed July 2026).